A. POST TENSIONING NOTES:

A1 POST-TENSIONED STRANDS, ANCHORAGES, DUCTS AND OTHER SYSTEM COMPONENTS SHALL CONFORM TO LATEST DBC, ACI 318. DETAILS OF ALL PROPOSED MATERIALS INCLUDING MATERIAL TEST CERTIFICATES SHALL BE SUBMITTED TO THE ENGINEER FOR APPROVAL.

A2 ALL STRANDS SHALL BE 12.9mm NOMINAL DIAMETER 7 WIRE STRESS RELIEVED SUPER GRADE, LOW RELAXATION STRAND WITH MINIMUM BREAKING LOAD = 186 kN U.N.O.

A3 UNBONDED TENDONS SHALL NOT BE USED.

A4 THE FINAL JACKING FORCE SHALL NOT EXCEED 80% OF THE BREAKING LOAD (149 kN FOR 12.9mm STRAND).

A5 PARAMETERS ALLOWED FOR IN THE DESIGN ARE

DUCT WOBBLE COEFFICIENT	β	=	0.025 rad./m.
FRICTION CURVATURE COEFFICIENT	μ	=	0.20
PULL IN ALLOWANCE		=	6.00 mm
DUCT DEPTH		=	19 mm
ELASTIC MODULUS		=	195kN/mm2
RELAXATION		=	5%

A6 PRESTRESSING DUCTS SHALL BE PLACED AND MAINTAINED TO THE PROFILES SPECIFIED TO WITHIN ±2 mm.

A7 THE DUCT SHALL FOLLOW A SMOOTH PARABOLIC PROFILE BETWEEN HIGH AND LOW POINTS MAKING ALLOWANCE FOR VARIABLE SLAB SOFFITS.

A8 THE DUCTS SHALL BE SUPPORTED AT THE CORRECT LEVELS ON APPROVED CHAIRS PLACED AT MAX. 1000 mm CENTRES.

A9 MINIMUM CLEAR COVER TO DUCTS

SLABS - 35mm
BEAMS - 50mm

A10 THE STRESSING SEQUENCE SHALL BE AS FOLLOWS:

STAGE 1:

25 % OF JACKING LOAD WHEN CONCRETE STRENGTH REACHES 9 N/Sq.mm.

STAGE 2:

100 % OF JACKING LOAD WHEN CONCRETE STRENGTH REACHES 25 N/Sq.mm. (TRANSFER STAGE).

A11 NOTE THAT THE CENTRAL STRAND WITHIN EACH DUCT SHOULD BE STRESSED FIRST, FOLLOWED BY THE NEXT STRAND ON EITHER SIDE.

A12 ON APPROVAL OF THE STRAND EXTENSIONS, THE DUCTS SHALL BE GROUTED. THE PHYSICAL AND CHEMICAL PROPERTIES OF THE GROUT SHOULD COMPLY WITH ACI 318.

A13 SHRINKAGE COMPENSATED GROUT SHALL BE USED SUCH THAT THE UNRESTRAINED EXPANSION IS 0 - 5%.

A14 ON COMPLETION OF THE GROUTING OPERATION, GROUT RECORDS SHALL BE SUBMITTED TO THE ENGINEER.

A15 RECESSES AROUND ANCHORS SHALL BE FILLED WITH A MATERIAL COMPATIBLE WITH THE CONCRETE STRENGTH APPROVED BY THE ENGINEER.

A16 CALCULATIONS FOR BACKPROPPING THE SLAB SHALL BE SUBMITTED TO THE ENGINEER FOR APPROVAL. PROPS SHALL NOT BE REMOVED UNTIL THE FINAL STRESS HAS BEEN APPLIED AND THE ENGINEER'S APPROVAL HAS BEEN OBTAINED.

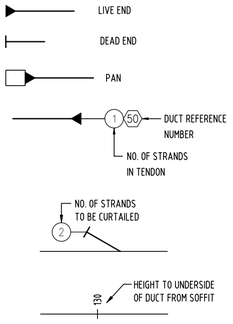
A17 THE SOFFIT OF THE SLAB SHALL BE INDELLIBLY MARKED IN RED DIRECTLY BELOW EACH POST-TENSIONING DUCT, ALONG THE ENTIRE DUCT LENGTH.

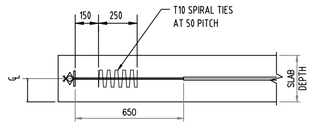
A18 SUBMIT FOR APPROVAL SHOP DRAWING TENDON DETAILS, TENDON LAYOUT, TENDON PROFILE, ANCHORAGE DETAILS, NON PRESTRESSED REINFORCEMENT, STRESSING SEQUENCES AND BLOCK OUT & FINISHING DETAILS

A19 MAXIMUM SPACING OF TENDON 7T (T=SLAB THICKNESS) AND AS PER ACI 318.

A20 PT SPECIALIST CONTRACTOR TO CO-ORDINATE THE DESIGN WITH ARCHITECTURE, STRUCTURE AND MEP DESIGN INTENT.

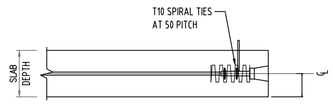
B. STANDARD SYMBOLS USED





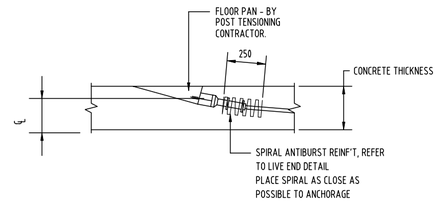
DEAD END ANCHORAGE
SCALE N.T.S.

℄ DIMENSION IS GIVEN ON STRESSING LAYOUT PLAN AND IS FROM THE SOFFIT OF THE SLAB TO THE CENTRELINE OF THE ANCHORAGES U.N.O.



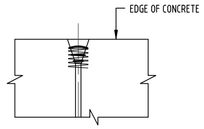
LIVE END ANCHORAGE.
SCALE N.T.S.

℄ DIMENSION IS GIVEN ON STRESSING LAYOUT PLAN AND IS FROM THE SOFFIT OF THE SLAB TO THE CENTRELINE OF THE ANCHORAGES U.N.O.



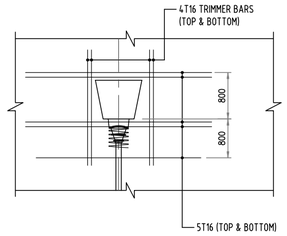
SLAB OR BAND LIVE END WITH PAN DETAIL
SCALE N.T.S.

℄ DIMENSION IS GIVEN ON STRESSING LAYOUT PLAN AND IS FROM THE SOFFIT OF THE SLAB TO THE CENTRELINE OF THE ANCHORAGES U.N.O.

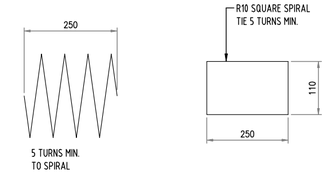


PLAN

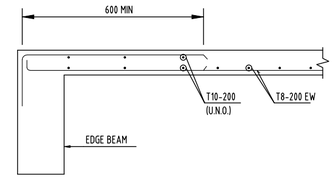
LIVE END ANCHORAGE.
SCALE N.T.S.



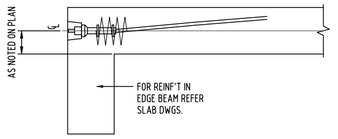
TYPICAL TRIMMING REINFORCEMENT AT STRESSING PANS
SCALE N.T.S.



TYPICAL DETAILS FOR ANTIBURST REINFORCEMENT
SCALE N.T.S.

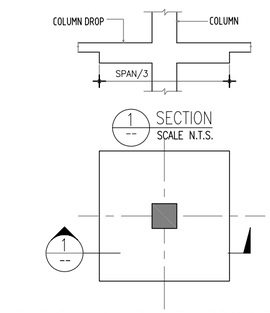


TYPICAL SLAB EDGE REINFORCEMENT DETAIL
SCALE N.T.S.

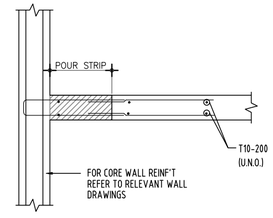


TYP REINFORCEMENT DETAIL FOR STRESSING PANS NEAR SLAB EDGES, BEAMS AND CORES
SCALE N.T.S.

TYPICAL BURSTING REINFORCEMENT
SCALE N.T.S.



TYPICAL SLAB WITH DROP DETAIL
SCALE N.T.S.



TYPICAL SLAB/CORE WALL JUNCTION
SCALE N.T.S.

NOTES.

1. THE DESIGN SHALL BE LIMIT STATE DESIGN IN ACCORDANCE WITH LATEST ACI 318,DBC AND CONCRETE SOCIETY TECHNICAL REPORT No. 43 AND SHALL ACCOUNT FOR THE FOLLOWING.
   - JACKING FORCE SHALL NOT EXCEED 75% OF THE CHARACTERISTIC STRENGTH OF THE STRAND.
   - PUNCHING SHEAR SHALL BE CHECKED AT COLUMN/WALL SUPPORTS.
   - LONG-TERM DEFLECTION SHALL NOT EXCEED SPAN/250
   - DEFLECTION AFTER THE INSTALLATION OF SUPPER IMPOSED DEAD LOADS (FINISHES, PARTITIONS ETC.) SHALL NOT EXCEED SPAN/500 (OR) 20mm, WHICHEVER IS LESSER.
   - THE EFFECT OF WALL RESTRAINT SHALL BE TAKEN INTO ACCOUNT IN THE DESIGNS PROVIDING ADDITIONAL REINFORCEMENT OR POUR STRIPS AS REQUIRED
   - MINIMUM PRE-STRESS AT ANY LOCATION SHALL BE 0.7 N/mm2.
   - SERVICEABILITY CLASSIFICATION MAY BE TAKEN AS CLASS 3.
   - ELONGATION REQUIREMENT OF THE STRANDS SHALL BE SUBMITTED.
2. POST-TENSIONED STRANDS, ANCHORAGES, DUCTS AND OTHER SYSTEM COMPONENTS SHALL CONFORM TO ACI, BS 5896 AND BS 4447. DETAILS OF ALL PROPOSED MATERIALS INCLUDING CERTIFICATES SHALL BE SUBMITTED TO THE ENGINEER FOR APPROVAL.
3. CONTRACTOR TO BE RESPONSIBLE TO OBTAIN STATUARY BODY APPROVAL FOR ALL DESIGNS AND DRAWINGS.
4. NORMAL COVER TO ALL REINFORCEMENT (INCLUDING LINKS) SHALL NOT BE LESS THAN 25mm FOR SLABS AND 30mm FOR BEAMS.
   ROOF POST TENSIONED SLAB SHALL ACHIEVE 4Hrs.FIRE RESISTANCE. ALL OTHER ELEMENTS SHALL ACHIEVE A 2 HOUR FIRE RESISTANCE.
5. THE STRESSING PROCEDURE SHALL FOLLOW GOOD ENGINEERING PRACTICE AND SHALL BE AS PER THE SPECIFICATION.
6. GROUT MATERIAL SHALL BE AS PER THE SPECIFICATION AND THE MIX DESIGN SHALL BE TO THE APPROVAL OF THE ENGINEER. MINIMUM COMPRESSIVE STRENGTH OF GROUT MEASURED ON A 150mm.CUBE SHALL BE:
   - 30 N/mm^2 AT 7 DAYS.
   - 40 N/mm^2 AT 28 DAYS.
7. SPACING AND PROVISION OF NON PRE STRESSED REINFORCEMENT AND TENDONS SHOULD BE SUFFICIENTLY ADEQUATE IN BOTH DIRECTION FOR THE AVOIDANCE OF ANY CRACK FORMATION FROM SHRINKAGE.
8. SLEEVES TO BE PROVIDED IN THE SLAB FOR FUTURE DEVELOPMENT. SIZES OF THE SLEEVES, LOCATION AND DETAILS AS SHOWN IN THE MEP/ARCHITECTURAL DRAWINGS.
9. A CONVENTIONAL REINFORCEMENT OF T10-200 EW-T&B (MINIMUM) SHALL BE PROVIDED IN ALL PT FLOORS.
10. THE DESIGN OF PT SLABS TO BE AS PER ACI 318 STANDARDS.
11. THE GROUTING OPERATION SHALL BE CARRIED OUT AS PER THE SPECIFICATIONS.
    A DETAILED METHOD STATEMENT SHALL BE SUBMITTED, WHICH COVERS ALL ASPECTS OF SITE WORKS INCLUDING:
    - ORGANISATION CHART SHOWING THE PROJECT TEAM
    - STRESSING OPERATIONS
    - GROUTING OPERATIONS
    - REMOVAL OF FORMWORK AND PROPS
    - ITP AND RISK ASSESSMENT
12. DETAILS ARE INDICATIVE AND THE MINIMUM REQUIRED CONTRACTOR TO PREPARE HIS WORKING DRAWING TO SUIT WITH THE SITE SPECIFIC REQUIREMENTS AND SUBMIT TO THE ENGINEERS FOR DESIGN CODES, LOCAL AUTHORITY.
13. PT SLAB SHALL BE DESIGNED TO RESIST IN TENSION 1% OF THE COLUMNS AXIAL LOAD TO PROVIDE LATERAL SUPPORT. COLUMN LOADS WILL BE ISSUED TO PT SPECIALIST UPON REQUEST.

NOTES :
1. THIS DRAWING MAY NOT BE REPRODUCED BY ANY MEANS WITHOUT PRIOR APPROVAL FROM EMSQUARE
2. DRAWINGS ARE NOT TO BE SCALED,STATED DIMENSIONS ONLY ARE TO BE TAKEN.
3. ANY INCONSISTENCY OR DISCRIPANCY IS TO BE BROUGHT TO THE ATTENTION OF & RESOLVED BY THE ARCHITECT/ENGINEER BEFORE THE WORK IS PUT IN HAND.

DATE	DESCRIPTION	REV.	BY
13.04.2026	FOR AUTHORITY APPROVAL	00	MS

CONSULTANT :

EMSQUARE
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OWNER :

G F S REAL ESTATE L.L.C

JOB TITLE :

PROPOSED RESIDENTIAL BUILDING (B+G+5+ROOF)

PLOT NO. / PARCEL ID	BLOCK / COMMUNITY
5310319	SAIH SHUAIB 2

DRAWING TITLE : STANDARD POST TENSION DETAILS

ARCHITECT :	STRUCTURAL ENGR. :	DRAWN :
SHAFI	Eng.Vincent	MUFADDAL
SCALE :	**DATE :**	**CHECKED BY :**
NTS	13.04.2026	Eng.Vincent
JOB NO :	**DRAWING NO. :**	**REVISION NO. :**
—	S-01_6	

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